

MMF062 Vehicle Motion Engineering

Design Task 3 - Group 5

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January 9, 2025

Task 1 - Quarter Car Model Transfer Functions in Matrix Form

Task 1.1: Model and derive equations

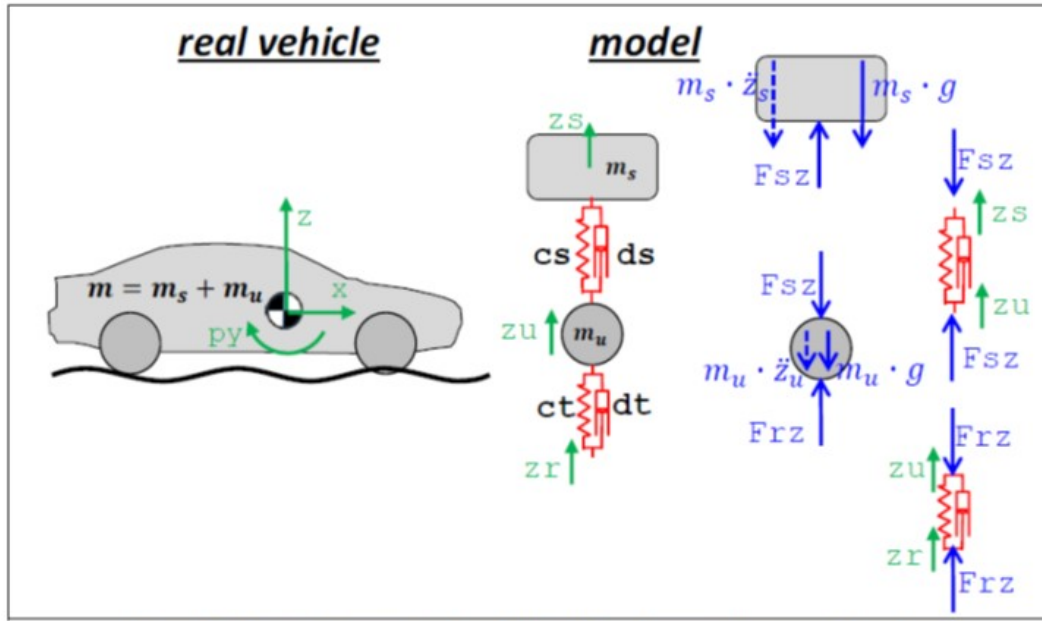


Figure 5-12: One-dimensional model with two dynamic degrees of freedom

Figure 1: Free Body Diagram

Mathematical model Equilibrium

$$m_s \ddot{z}_s = F_{sz} - m_s g \quad (1)$$

$$m_u \ddot{z}_u = F_{rz} - m_u g - F_{sz} \quad (2)$$

Displacement measured from the static equilibrium

$$F_{sz} = c_s(z_u - z_s) + d_s(\dot{z}_u - \dot{z}_s) + m_s g \quad (3)$$

$$F_{rz} = c_t(z_r - z_u) + d_t(\dot{z}_r - \dot{z}_u) + (m_s + m_u)g \quad (4)$$

Excitation:

$$z = z(t) \quad (5)$$

Solve for $\ddot{z}_s$ and $\ddot{z}_u$

$$\ddot{z}_s = \frac{c_s(z_u - z_s) + d_s(\dot{z}_u - \dot{z}_s)}{m_s} \quad (6)$$

$$\ddot{z}_u = \frac{c_t(z_r - z_u) + d_t(\dot{z}_r - \dot{z}_u) - c_s(z_u - z_s) - d_s(\dot{z}_u - \dot{z}_s)}{m_u} \quad (7)$$

The equation of motion in the state-space form

$$\dot{x} = Ax + Bu \quad (8)$$

X is a state vector

$$\mathbf{x} = \begin{bmatrix} z_s \\ z_u \\ \dot{z}_s \\ \dot{z}_u \end{bmatrix}$$

The state-space form of $\dot{x} = Ax + Bu$

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu$$

$$\mathbf{x} = \begin{bmatrix} \dot{z}_s \\ \dot{z}_u \\ \ddot{z}_s \\ \ddot{z}_u \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{c_s}{m_s} & \frac{c_s}{m_s} & -\frac{d_s}{m_s} & \frac{d_s}{m_s} \\ \frac{c_s}{m_u} & -\frac{(c_s+c_t)}{m_u} & \frac{d_s}{m_u} & -\frac{(d_s+d_t)}{m_u} \end{bmatrix} \begin{bmatrix} z_s \\ z_u \\ \dot{z}_s \\ \dot{z}_u \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{c_t}{m_u} \end{bmatrix} z_r$$

Task 1.2: Determine transfer function matrix $\mathbf{H}(\omega)$

. The input and the output equation

$$\dot{x} = Ax + Bu \quad (9)$$

$$y = Cx + Du \quad (10)$$

Using Laplace Transformation

$$sX(s) = AX(s) + BU(s) \quad (11)$$

$$Y(s) = CX(s) + DU(s) \quad (12)$$

From equation 11

$$(sI - A)X(s) = BU(s) \quad (13)$$

$$X(s) = [(sI - A)^{-1}]BU(s) \quad (14)$$

Substituting equation (14) in equation (11)

$$Y(s) = [C(sI - A)^{-1}BU(s)] + DU(s) \quad (15)$$

$$Y(s) = [C(sI - A)^{-1}B + D]U(s) \quad (16)$$

$$H(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D \quad (17)$$

Task 1.3: Plot transfer functions $H(\omega)$

. The B, C and D matrices for corresponding transfer function for each single front and rear wheel

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{c_t}{m_u} \end{bmatrix}$$

1. $H(\omega)zr \rightarrow \ddot{z}_s$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad D = 0$$

2. $H(\omega)zr \rightarrow (zu - zs)$

$$\mathbf{C} = \begin{bmatrix} -1 & 1 & 0 & 0 \end{bmatrix} \quad D = 0$$

3. $H(\omega)zr \rightarrow (\delta F_{rz})$

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \quad D = 0$$

Plot the magnitude of the three transfer functions versus frequencies from 0.1 to 50 Hz. Refer below figures.

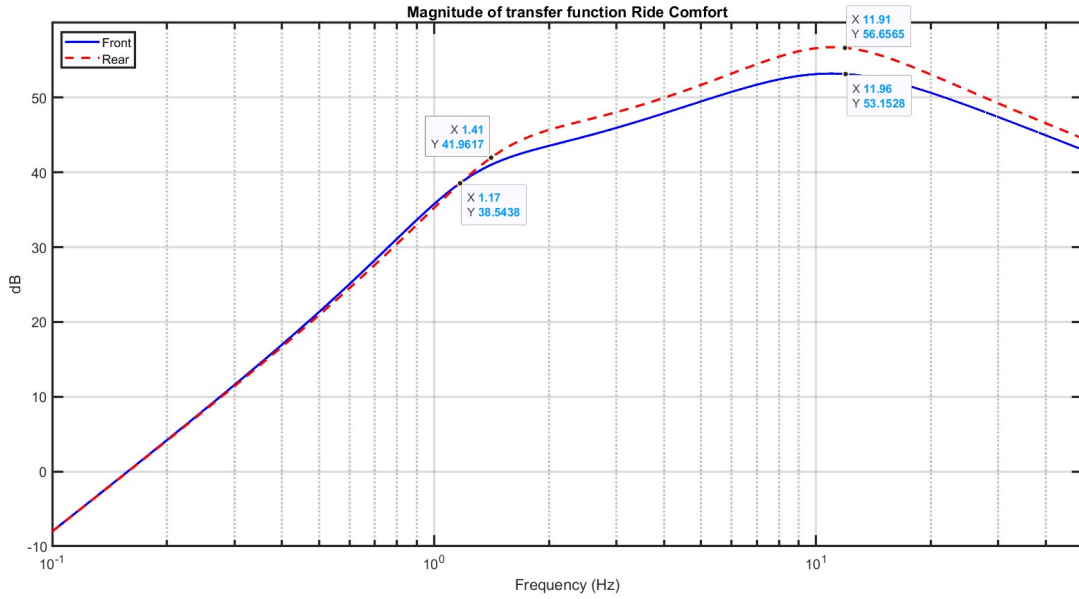


Figure 2: Ride comfort $H(\omega)zr \rightarrow \ddot{z}_s$

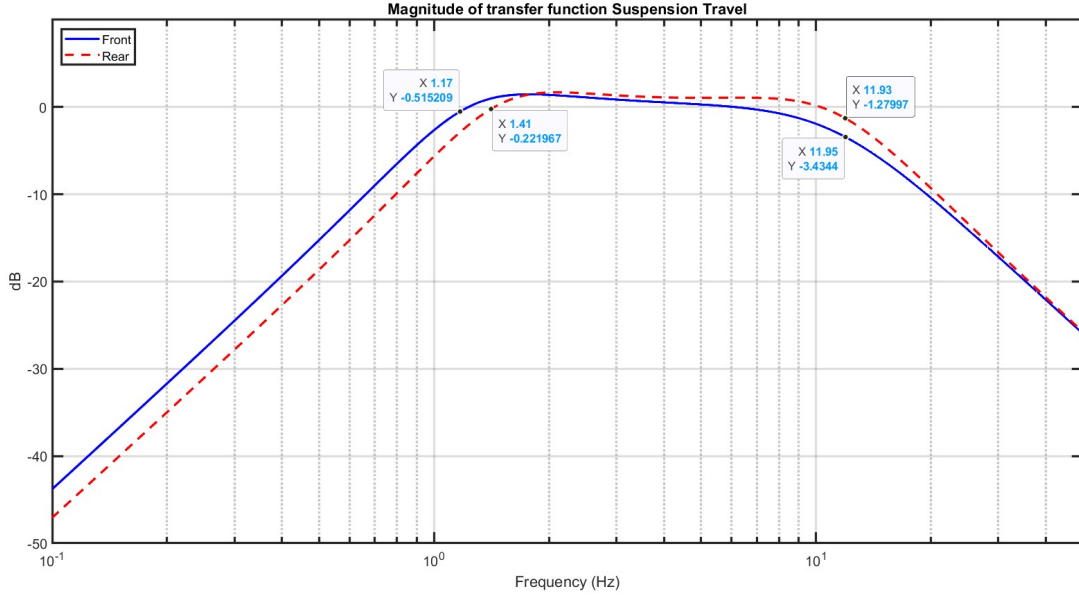


Figure 3: suspension travel $H(\omega)zr \rightarrow (zu - zs)$

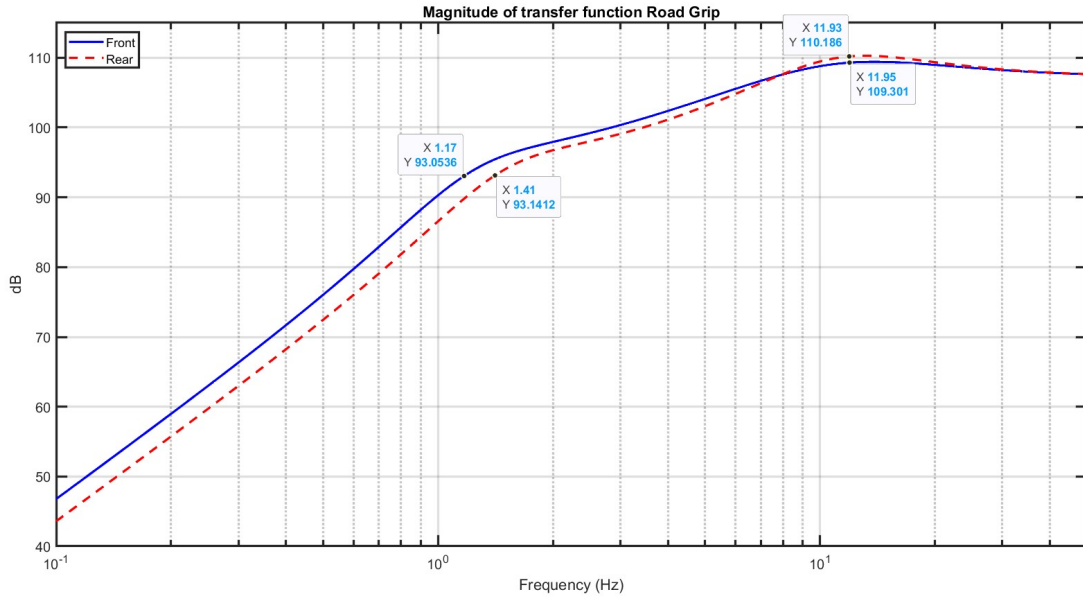


Figure 4: Tyre force $H(\omega)zr \rightarrow (\delta F_{rz})$

Task 1.4: Identify natural frequencies

. The natural frequency for front and rear wheels are calculated using the expression from the compendium. The natural frequency for Bounce and Wheel Hop is given below.

$$\omega_{\text{Bounce}} = \sqrt{\frac{1/(\frac{1}{c_s} + \frac{1}{c_t})}{m_s}} \quad \omega_{\text{WheelHop}} = \sqrt{\frac{c_s + c_t}{m_u}}$$

Wheels	Resonance frequency Bounce	Resonance frequency WheelHop
Front	1.1701	11.9514
Rear	1.4145	11.9307

Table 1: Resonance Frequency of Bounce and WheelHop

The frequency of the bounce for the front and the rear wheels when we compare it with the Task 1.3 graphs the values obtained in the graph almost matches with the natural frequency. We can see that the frequency bounce at the rear is greater then the front because the sprung mass at the front is larger and lower bounce frequency at the front provides more comfort to the passenger. The suspension stiffness and the frequency bounce at the rear is higher to provide load stability.

The frequency of the WheelHop for the front and rear wheels matches with the values obtained in the Task 1.3 graphs. The weight distribution is biased slightly towards the front and the front wheel has slightly lower unsprung mass compared to the rear and the suspension at the front is slightly stiffer because of this frequency wheelhop is slightly higher at the front.

Task 2 - Study of the suspension stiffness and damping

Task 2.1: Plot response spectrum and calculate rms values

The response spectra and the rms values can be calculated using the below equation.

$$G(\omega) = |H(\omega)|^2 \cdot G_{z_r}(\omega)$$

$$rms = \sqrt{\int_0^\infty G(\omega) d\omega} \approx \sqrt{\sum_i^N G(\omega_i) \cdot \Delta\omega}$$

Plot the response spectra for the Acceleration of the sprung mass and the Tyre force. Refer below figures.

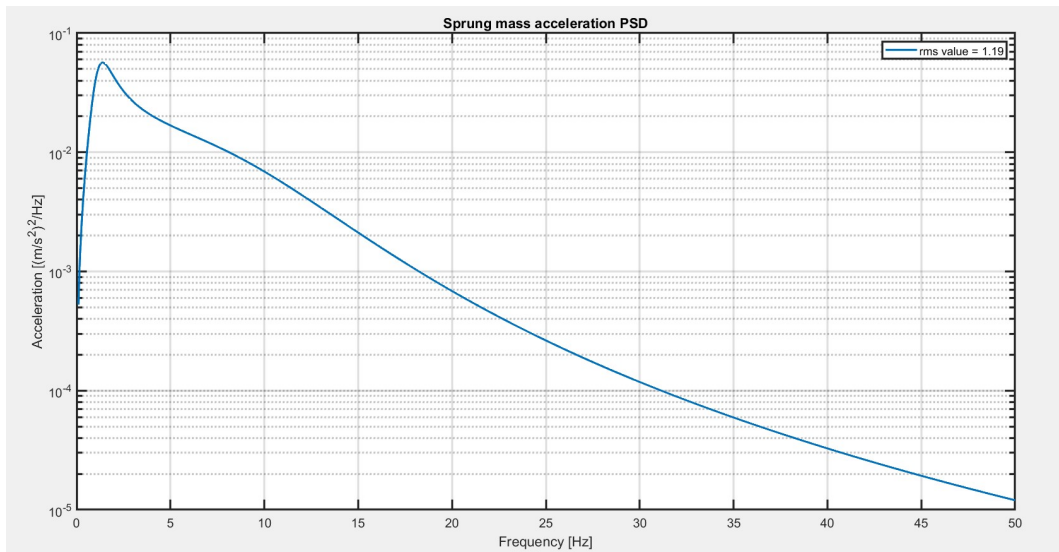


Figure 5: Acceleration of the sprung mass vs Frequency

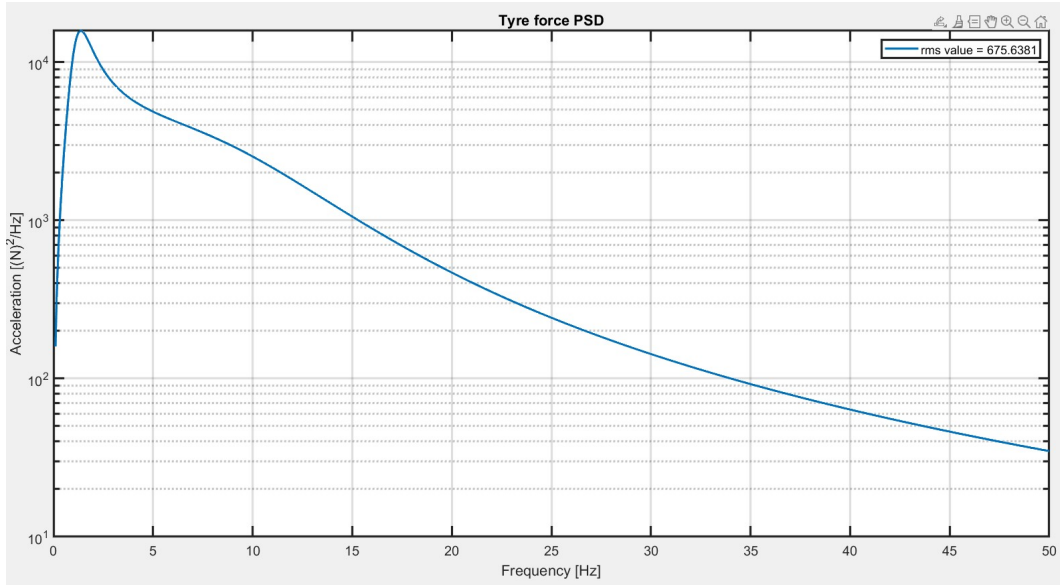


Figure 6: Tyre force vs Frequency

Task 2.2: Balance Ride comfort and Tyre force/road grip

- 1: For each stiffness value $[0.5, 0.75, 1, 1.25, 1.5, 2] \times \text{frontWheelSuspStiff}$ vary the damping according to $[1000 : 100 : 9000]$. The rms values for the sprung mass acceleration and the tyre force is calculated from the equation mentioned in the Task 2.1.
- 2: Plot the rms values for the sprung mass acceleration vs the damping coefficient $[1000 : 100 : 9000]$ and rms values for the tyre force vs. the damping coefficient $[1000 : 100 : 9000]$.

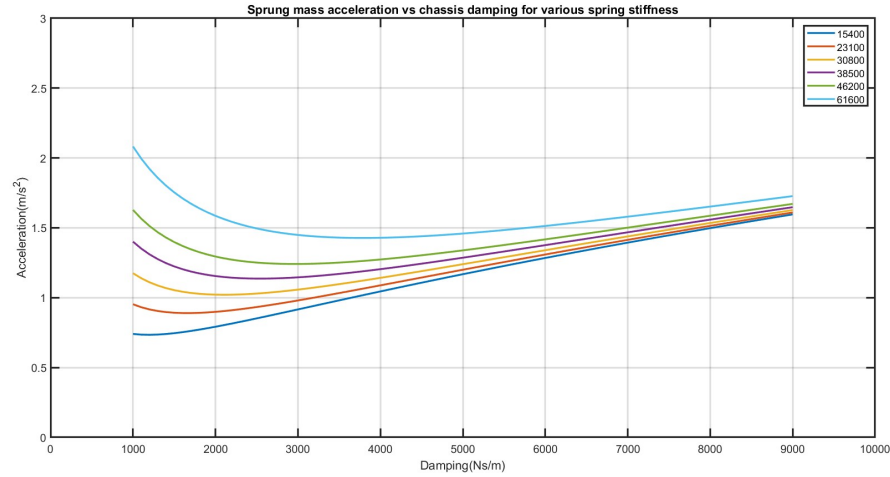


Figure 7: Sprung mass acceleration vs Damping coefficient

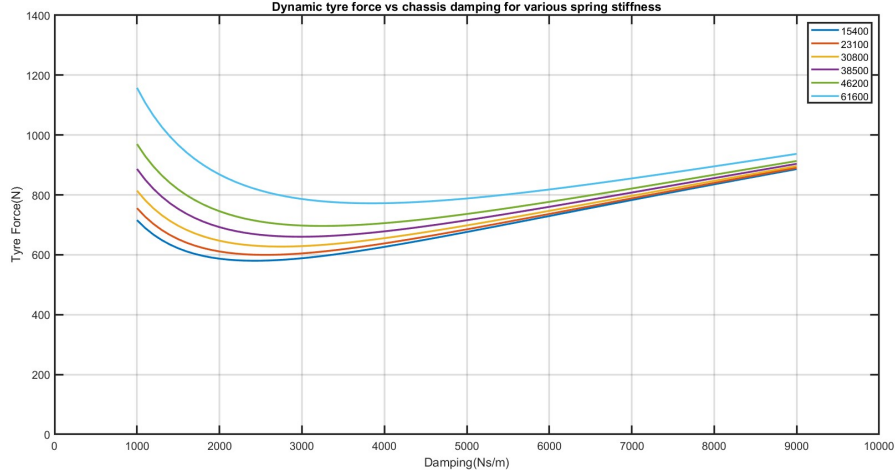


Figure 8: Tyre force vs Damping coefficient

Higher spring stiffness increases the sprung mass acceleration which gives bumpier ride for the driver. Ride comfort is favored by the softer spring stiffness and higher damping values within the optimal range.

Higher spring stiffness reduces tyre force which means better road grip and handling. Road grip is favored by stiffer spring stiffness and moderate damping values within the optimal range.

- 3. Plot the identified optimal damping values vs spring stiffness values ($[0.5, 0.75, 1, 1.25, 1.5, 2] \times \text{frontWheelSuspStiff}$).

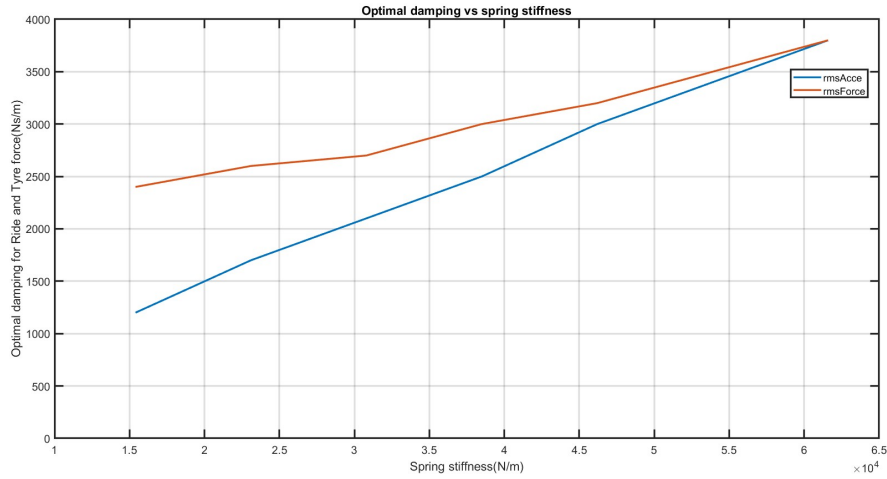


Figure 9: Optimal damping values vs spring stiffness values

From the figure 9 we can say that Sprung mass acceleration i.e Ride comfort requires softer damping because excessive damping transfers the road irregularities to the vehicle body which means reduces the ride comfort. Tyre Force i.e Road grip requires higher damping because it reduces oscillation of the unsprung mass and helps to maintain contact between the tyre and the road which means improves the road grip.

Task 3 - Ride comfort and the use of ISO-2631

Task 3.1: Calculation of daily whole-body exposure values

The whole-body vibration exposure value is calculated when driving on the specified road in 110 km/h during an eight hour period and compared the calculate values with the limit specified in Directive 2002/44/EC.

- 1: Sprung mass acceleration response spectra is calculate for the different road types. Refer the matlab file Task3 Skeleton.
- 2: ISO-filtered sprung mass acceleration rms values is calculated for the different road types which is mentioned below.
 - Weighted Rms Acceleration Smooth = 0.3940
 - Weighted Rms Acceleration Rough = 1.2247
 - Weighted Rms Acceleration Very Rough = 4.0286.
- 3: The time averaged vibration exposure value is calculated using the below mentioned equation.

$$a_{w,av} = \sqrt{\frac{\sum_i a_{wi}^2 \cdot T_i}{\sum_i T_i}};$$

timeWeightedRmsAcceleration = 36.5802 m/s^2 . Number of trips per day is 5.5.

Task 3.2: Modification of vehicle velocity

The vehicle speed is modified individually on the three different road types so that the vibration exposure limit is not exceeded on each road type. The selected vehicle speed is specified and the weighted Rms Acceleration is noted for each road condition and it is below 1.15 m/s^2 acceleration. Refer below table 2 for the results.

Road condition	Vehicle speed(km/hr)	Weighted Rms acceleration (m/s^2)
Smooth	110	0.3940
Rough	101	1.1488
Very Rough	8.5	1.1199

Table 2: Modification of vehicle velocity

timeWeightedRmsAcceleration = 51.6519 m/s^2 , Number of trips per day is 2.1928.